

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281869

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

ORIHARA; Maho et al.

REACTOR AND METHOD FOR PRODUCING SAME, GAS RECOVERY DEVICE, AND GAS RECOVERY SYSTEM

Abstract

A reactor includes a honeycomb structure including: an outer peripheral wall; porous partition walls provided on an inner side of the outer peripheral wall, the porous partition walls defining first cells, second cells, and third cells through which a process gas containing a trapping target gas, each of the first cells, the second cells, and the third cells extending from an inflow end face to an outflow end face of the honeycomb structure; first plugged portions provided at the first cells on the inflow end face side; second plugged portions provided at the second cells on the outflow end face side; and third plugged portions provided at the third cells on the outflow end face side, the third cells being interposed between the first cells and the second cells, and a functional material having a pellet shape, the functional material being filled in the third cells.

Inventors: ORIHARA; Maho (Toyohashi-Shi, JP), WADA; Takuya (Konan-Shi, JP), MURAKAMI; Keiya (Obu-Shi, JP), OYABU; Tomofumi (Nagoya-Shi, JP), TAKAGI; Yoshifumi (Chiryu-Shi, JP), MIYAIRI; Yukio (Nagoya-Shi, JP), AOKI; Yoichi (Yatomi-Shi, JP)

Applicant: NGK INSULATORS, LTD. (Nagoya-City, JP)

Family ID: 1000008670028

Assignee: NGK INSULATORS, LTD. (Nagoya-City, JP)

Appl. No.: 19/213095

Filed: May 20, 2025

Foreign Application Priority Data

WO

PCT/JP2022/046958

Dec. 20, 2022

Related U.S. Application Data

Publication Classification

Int. Cl.: **B01D53/04** (20060101); **B01D53/96** (20060101)

U.S. Cl.:

CPC **B01D53/0438** (20130101); **B01D53/96** (20130101); B01D2253/304 (20130101); B01D2253/308 (20130101); B01D2259/40086 (20130101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of priorities to PCT Patent Application No. PCT/JP2022/046958 filed on Dec. 20, 2022, the entire contents of which are incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] This invention relates to a reactor and a method for producing the same, a gas recovery device, and a gas recovery system.

BACKGROUND OF THE INVENTION

[0003] As a measure against global warming, there is a growing demand for the effective use of CO₂, including the recovery of CO₂ from factory exhaust gases, and combustion exhaust gases of CO₂ emission sources such as thermal power plants, and direct CO₂ recovery (direct air capture: DAC) and fixation from the atmosphere, as well as methanation of recovered CO₂.

[0004] The proposed main method for recovering CO₂ is to adsorb CO₂ onto an adsorbent capable of adsorbing CO₂, desorb the CO₂ by changing a temperature, pressure, humidity, and the like, recover it as highly concentrated CO₂, and use it as a raw material for the chemical industry, or to inject it into the underground and fix it.

[0005] The adsorbent is supported on porous pellets, porous particles, fiber filters, and honeycomb structures and used (e.g., Non-Patent Literature 1).

[0006] Non-Patent Literature 1 describes a reactor in which an adsorbent is supported (an adsorbent layer is formed) on each surface of partition walls of a honeycomb structure made of a mullite material, with a thickness of partition walls of 0.15 mm, a cell density of 400 cells/inch² (62 cells/cm²) and an opening ratio of 75.0%. Since an amount of CO₂ recovered is proportional to an amount of the adsorbent retained in the reactor, it is sufficient to place as many adsorbents as possible in the reactor.

[0007] However, in the method for supporting the adsorbent on each surface of the partition walls of the honeycomb structure, increasing the amount of the adsorbent supported causes the opening ratio of the honeycomb structure to decrease due to the thickness of the adsorbent layer, so that the pressure loss increases when the process gas containing CO₂ is circulated. As a result, a kinetic energy required for allowing the process gas to flow through the honeycomb structure increases, resulting in an increased reactor operating cost. As the thickness of the adsorbent layer increases, it becomes difficult for the process gas to reach the interior of the adsorbent layer sufficiently, so that adsorption performance of CO₂ also decreases. Similarly, in the CO₂ desorption process, it is difficult to desorb CO₂ sufficiently from the interior of the adsorbent layer.

[0008] In view of deterioration of the adsorbent when the reactor is used for a long period of time,

it is effective to retain as much adsorbents as possible in the reactor to extend the maintenance period, but the above method has limitations in increasing the amount of the adsorbent retained. [0009] On the other hand, the methods for supporting the adsorbent on porous pellets and forming the adsorbent itself into pellets can be considered, but these methods are inferior to the method for supporting the adsorbent on each surface of the partition walls of the honeycomb structure in terms of contact area with the process gas and of pressure loss. It can also be considered that the partition walls of the honeycomb structure are formed with the adsorbent, but, in this case, it is difficult for the process gas to diffuse sufficiently in the direction of the thickness of the partition walls, and the strength of the honeycomb structure is also reduced.

[0010] Although the case of the reactor in which the trapping target gas is CO.sub.2 and the adsorbent capable of adsorbing CO.sub.2 is used is described as an example, the same problem as described above may occur for a reactor in which the trapping target gas is a component other than CO.sub.2 and a functional material other than the adsorbent capable of adsorbing CO.sub.2 is used.

[0011] This invention was made to solve the above problems, and an object of this invention is to provide a reactor that can increase an amount of a trapping target gas by increasing an amount of a functional material retained while suppressing an increase in pressure loss, and a method for producing the same, a gas recovery device, and a gas recovery system.

PRIOR ART

Non-Patent Literature

[0012] [Non-Patent Literature 1] “Cost and Evaluation of Direct Air Capture (DAC) Method for Carbon Dioxide (Vol. 2)-Adsorption Separation Process-”, Center for Low Carbon Society Strategy, Japan Science and Technology Agency, March 2021.

SUMMARY OF THE INVENTION

[0013] As results of intensive studies for reactors including honeycomb structures, the inventors have found that the above problems can be solved by selectively filling specific cells of a honeycomb structure having a specific structure with a functional material, and they have completed this invention. In other words, the invention is exemplified as follows:

[0014] (1) A reactor, comprising: [0015] a honeycomb structure comprising: [0016] an outer peripheral wall; [0017] porous partition walls provided on an inner side of the outer peripheral wall, the partition walls defining first cells, second cells and third cells through which a process gas containing a trapping target gas can flow, each of the first cells, the second cells and the third cells extending from an inflow end face to an outflow end face of the honeycomb structure; [0018] first plugged portions provided at the first cells on the inflow end face side; [0019] second plugged portions provided at the second cells on the outflow end face side; and [0020] third plugged portions provided at the third cells on the outflow end face side, the third cells being interposed between the first cells and the second cells; and [0021] a functional material having a pellet shape which has a particle diameter of 0.025 to 10 mm, the functional material being filled in the third cells.

[0022] (2) The reactor according to (1), wherein the honeycomb structure further comprises fourth plugged portions provided at the third cells on the inflow end face side.

[0023] (3) The reactor according to (1) or (2), wherein the third plugged portions are porous.

[0024] (4) The reactor according to any one of (1) to (3), wherein the first plugged portions and the second plugged portions are dense.

[0025] (5) The reactor according to any one of (2) to (4), wherein the fourth plugged portions are porous.

[0026] (6) The reactor according to any one of (1) to (5), wherein each of the first cells, the second cells, and the third cells has a triangular, quadrangular, hexagonal, or octagonal shape, or a combination thereof, in a cross section orthogonal to an extending direction of the first cells, the second cells, and the third cells.

[0027] (7) The reactor according to any one of (1) to (6), wherein the outer peripheral wall and the

partition walls comprise at least one selected from cordierite, mullite, alumina, silicon carbide, and Si-bonded silicon carbide as a main component.

[0028] (8) The reactor according to any one of (1) to (7), wherein the partition walls have a thickness of 0.05 mm to 5 mm.

[0029] (9) The reactor according to any one of (1) to (8), wherein the partition walls have a porosity of 30% or more and less than 80%.

[0030] (10) The reactor according to any one of (1) to (9), wherein the partition walls have an average pore diameter of 10 μm to 300 μm .

[0031] (11) The reactor according to any one of (1) to (10), wherein the functional material is an adsorbent.

[0032] (12) The reactor according to (11), wherein the adsorbent is at least one selected from an amine compound, an organometallic complex, and a nanoporous ceramic or mesoporous silica on which the amine compound and/or the organometallic complex are supported.

[0033] (13) The reactor according to (12), wherein a particle diameter of the functional material having the pellet shape is lower on the outflow end face side than in a central portion in the extending direction of the third cells.

[0034] (14) The reactor according to (12) or (13), wherein the particle diameter of the functional material having the pellet shape is higher on the outer peripheral wall side than in the central portion in a direction orthogonal to the extension direction of the third cells.

[0035] (15) The reactor according to any one of (1) to (14), wherein the first cells and/or the second cells are located at positions facing the outer peripheral wall.

[0036] (16) The reactor according to any one of (1) to (15), wherein the first plugged portions, the second plugged portions, and the third plugged portions are made of a resin sheet.

[0037] (17) The reactor according to any one of (2) to (16), wherein the third plugged portions and the fourth plugged portions are made of a porous resin sheet.

[0038] (18) The reactor according to any one of (1) to (17), wherein the honeycomb structure has a quadrangular pillar shape having a length of one side of the inflow end face and the outflow end face of 100 to 500 mm, and a length in the extending direction of the first cells, the second cells, and the third cells of 100 to 1,000 mm.

[0039] (19) A method for producing a reactor, the method comprising the steps of: [0040] preparing a honeycomb structure comprising: an outer peripheral wall and porous partition walls provided on an inner side of the outer peripheral wall, the porous partition walls defining first cells, second cells, and third cells through which a process gas containing a trapping target gas can flow, each of the first cells, the second cells and the third cells extending from an inflow end face to an outflow end face of the honeycomb structure, the third cells being interposed between the first cells and the second cells; [0041] forming second plugged portions and third plugged portions on the outflow end face side of the second cells and the third cells, respectively; [0042] filling the third cells with a functional material having a pellet shape which has a particle diameter of 0.025 to 10 mm; and [0043] forming first plugged portions on the inflow end face side of the first cells.

[0044] (20) A method for producing a reactor, the method comprising the steps of: [0045] preparing a honeycomb structure comprising: an outer peripheral wall and porous partition walls provided on an inner side of the outer peripheral wall, the porous partition walls defining first cells, second cells, and third cells through which a process gas containing a trapping target gas can flow, each of the first cells, the second cells and the third cells extending from an inflow end face to an outflow end face, the third cells being interposed between the first cells and the second cells; [0046] forming second plugged portions and third plugged portions on the outflow end face side of the second cells and the third cells, respectively; [0047] filling the third cells with a functional material having a pellet shape which has a particle diameter of 0.025 to 10 mm; and [0048] forming first plugged portions and fourth plugged portions on the inflow end faces of the first cells and the third cells, respectively.

[0049] (21) A gas recovery device for recovering and releasing a trapping target gas contained in a process gas, the gas recovery device comprising: [0050] the reactor according to any one of (1) to (18); [0051] a gas feed pipe configured to feed the process gas or a purge gas to an inflow port of the reactor; and [0052] a gas discharge pipe configured to discharge the process gas or the purge gas from an outflow port of the reactor.

[0053] (22) The gas recovery device according to (21), [0054] wherein the gas feed pipe has a gas feed branch pipe that branches into two parts, the gas feed branch pipe being a first gas feed branch pipe configured to feed the process gas and a second gas feed branch pipe configured to feed the purge gas, [0055] wherein the gas discharge pipe has a gas discharge branch pipe that branches into two parts, the gas discharge branch pipe being a first gas discharge branch pipe configured to discharge the process gas and a second gas discharge branch pipe configured to discharge the purge gas, and [0056] wherein the gas recovery device further comprises: a feed gas switching valve configured to shut off the first gas feed branch pipe or the second gas feed branch pipe; and a discharge gas switching valve configured to shut off the first gas discharge branch pipe or the second gas discharge branch pipe.

[0057] (23) The gas recovery device according to (21), [0058] wherein the gas feed pipe has two independent gas feed pipes, the gas feed pipes being a first gas feed pipe configured to feed the process gas and a second gas feed pipe configured to feed the purge gas, [0059] wherein the gas discharge pipe has two independent gas discharge pipes, the gas discharge pipes being a first gas discharge pipe configured to discharge the process gas and a second gas discharge pipe configured to discharge the purge gas, [0060] wherein the reactor can be provided between the first gas feed pipe and the first gas discharge pipe or between the second gas feed pipe and the second gas discharge pipe, and [0061] wherein the gas recovery device further comprises a transfer mechanism configured to transfer the reactor between the first gas feed pipe and the first gas discharge pipe or between the second gas feed pipe and the second gas discharge pipe.

[0062] (24) The gas recovery device according to any one of (21) to (23), further comprising a heating mechanism configured to heat the reactor.

[0063] (25) A gas recovery system for recovering and releasing a trapping target gas contained in a process gas, the gas recovery system comprising: [0064] a gas recovery device comprising an attachable/detachable portion configured to attach and detach the reactor according to any one of (1) to (18), a gas feed pipe configured to feed the process gas to an inflow port of the reactor, and a gas discharge pipe configured to discharge the process gas from an outflow port of the reactor; [0065] a gas release device comprising an attachable/detachable portion configured to attach and detach the reactor, a gas feed pipe configured to feed a purge gas to the inflow port of the reactor, and a gas discharge pipe configured to discharge the purge gas from the outflow port of the reactor, and [0066] a transfer device configured to transfer the reactor that has recovered the trapping target gas by the gas recovery device to the gas release device and also to transfer the reactor that has released the trapping target gas by the gas release device to the gas recovery device.

[0067] (26) The gas recovery system according to (25), wherein the transfer device comprises a vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0068] FIG. 1A is a schematic view of an inflow end face of a reactor according to an embodiment of the invention;

[0069] FIG. 1B is a schematic view of the outflow end portion of the reactor in FIG. 1A;

[0070] FIG. 1C is a schematic view of the cross sectional taken along the line a-a' in FIGS. 1A and 1B;

[0071] FIG. 1D is a partially enlarged view for explaining the flow of a processing gas;
[0072] FIG. 2A is a schematic view of an inflow end face of a reactor according to another embodiment of the invention;
[0073] FIG. 2B is a schematic view of the outflow end face of the reactor in FIG. 2A;
[0074] FIG. 3 is a partially enlarged view of the inflow end face for illustrating the shape of each cell;
[0075] FIG. 4 is a partially enlarged view of the inflow end face for illustrating the shape of each cell;
[0076] FIG. 5 is a partially enlarged view of the inflow end face for illustrating the shape of each cell;
[0077] FIG. 6 is a partially enlarged view of the inflow end face for illustrating the shape of each cell;
[0078] FIG. 7 is a partially enlarged view of the inflow end face for illustrating the shape of each cell;
[0079] FIG. 8 is a partially enlarged view of the inflow end face for illustrating the shape of each cell;
[0080] FIG. 9A is a schematic view of an inflow end face of a reactor according to another embodiment of the invention;
[0081] FIG. 9B is a schematic view of the outflow end face of the reactor in FIG. 9A;
[0082] FIG. 9C is a schematic view of the cross sectional taken along the line a-a' in FIGS. 9A and 9B;
[0083] FIG. 10 is a partially enlarged view for illustrating the flow of a process gas;
[0084] FIG. 11A is a schematic view of an inflow end face of a reactor according to another embodiment of the invention;
[0085] FIG. 11B is a schematic view of the cross sectional taken along the line a-a' in FIG. 11A;
[0086] FIG. 12 is a schematic view of a structure of a gas recovery device according to an embodiment of the invention;
[0087] FIG. 13: a schematic view of a structure of a gas recovery device according to another embodiment of the invention; and
[0088] FIG. 14 is a schematic view of a structure of a gas recovery system according to an embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

[0089] A reactor according to an embodiment of the invention includes: a honeycomb structure comprising: an outer peripheral wall; porous partition walls provided on an inner side of the outer peripheral wall, the partition walls defining first cells, second cells and third cells through which a process gas containing a trapping target gas can flow, each of the first cells, the second cells and the third cells extending from an inflow end face to an outflow end face of the honeycomb structure; first plugged portions provided at the first cells on the inflow end face side; second plugged portions provided at the second cells on the outflow end face side; and third plugged portions provided at the third cells on the outflow end face side, the third cells being interposed between the first cells and the second cells; and a functional material having a pellet shape, the functional material being filled in the third cells.

[0090] A method for producing a reactor according to an embodiment of the invention includes the steps of: preparing a honeycomb structure including: an outer peripheral wall and porous partition walls provided on an inner side of the outer peripheral wall, the porous partition walls defining first cells, second cells, and third cells through which a process gas containing a trapping target gas can flow, each of the first cells, the second cells and the third cells extending from an inflow end face to an outflow end face of the honeycomb structure, the third cells being interposed between the first cells and the second cells; forming second plugged portions and third plugged portions on the outflow end face side of the second cells and the third cells, respectively; filling the third cells with

a functional material having a pellet shape; and forming first plugged portions on the inflow end face side of the first cells.

[0091] Also, a gas recovery device according to an embodiment of the invention is for recovering and releasing a trapping target gas contained in a process gas, the gas recovery system including: the reactor; a gas feed pipe configured to feed the process gas or a purge gas to an inflow port of the reactor; and a gas discharge pipe configured to discharge the process gas or the purge gas from an outflow port of the reactor.

[0092] Furthermore, a gas recovery system according to an embodiment of the invention is for recovering and releasing a trapping target gas contained in a process gas, the gas recovery system including: a gas recovery device including an attachable/detachable portion configure to attach and detach the reactor, a gas feed pipe configured to feed the process gas to an inflow port of the reactor, and a gas discharge pipe configured to discharge the process gas from an outflow port of the reactor; a gas release device including an attachable/detachable portion configure to attach and detach the reactor, a gas feed pipe configured to feed a purge gas to the inflow port of the reactor, and a gas discharge pipe configured to discharge the purge gas from the outflow port of the reactor, and a transfer device configured to transfer the reactor that has recovered the trapping target gas by the gas recovery device to the gas release device and also to transfer the reactor that has released the trapping target gas by the gas release device to the gas recovery device.

[0093] The reactor and the method for producing the same, the gas recovery device, and the gas recovery system according to an embodiment of the invention can increase an amount of a trapping target gas by increasing an amount of a functional material retained while reducing an increase in pressure loss, according to the above structure. The suppression of the increase in pressure loss can also lead to decreased operating costs of the reactor because kinetic energy required to allow the process gas to flow through the honeycomb structure can be reduced.

[0094] Hereinafter, embodiments of the invention will be specifically described with reference to the drawings. It should be understood that the invention is not limited to the following embodiments, and those which have appropriately added changes, improvements and the like to the following embodiments based on knowledge of a person skilled in the art without departing from the spirit of the invention fall within the scope of the invention.

<Reactor>

[0095] The reactor according to an embodiment of the invention can be suitably used to recover the trapping target gas contained in the process gas. Non-limiting examples of the process gas include exhaust gases emitted from factories, power plants, and the like, and the atmosphere. Non-limiting examples of the exhaust gas include combustion exhaust gases generated when burning fossil fuels, coal gasification gases made by gasifying coal or natural gases in thermal power plants, steel mills, and other facilities. Non-limiting examples of the trapping target gas include carbon dioxide (CO.sub.2), nitrogen oxides (NO.sub.x), sulfur oxides (SO.sub.x), hydrogen sulfide (H.sub.2S) and the like. Among these, the reactor according to an embodiment of the invention is particularly useful for recovering carbon dioxide (CO.sub.2) from combustion exhaust gases and the atmosphere.

[0096] FIG. 1A is a schematic view of an inflow end face of a reactor according to an embodiment of the invention. FIG. 1B is a schematic view of the outflow end portion of the reactor in FIG. 1A. FIG. 1C is a schematic view of the cross sectional taken along the line a-a' in FIGS. 1A and 1B.

[0097] As shown in FIGS. 1A to 1C, a reactor **100** includes a honeycomb structure including: an outer peripheral wall **10**; porous partition walls **30** provided on an inner side of the outer peripheral wall **10**, the porous partition walls **30** defining first cells **23**, second cells **24**, and third cells **25** each extending from an inflow end face **20** to an outflow end face **21** of the honeycomb structure; first plugged portions **40** provided at the first cells **23** on the inflow end face **20** side; second plugged portions **41** provided at the second cells **24** on the outflow end face **21** side; and third plugged portions **42** provided at the third cells **25** on the outflow end face **21** side, the third cells **25** being

interposed between the first cells **23** and the second cells **24**, and a functional material **50** filled in the third cells **25**. The first cells **23**, the second cells **24**, and the third cells **25** can allow the process gas containing the trapping target gas to flow therethrough.

[0098] Here, FIG. **1D** shows a partially enlarged view of the same cross section as FIG. **1C** to illustrate the flow of the process gas. It should be noted that in FIG. **1D**, the length in the extending direction of each cell is shorter than that in FIG. **1C** from the viewpoint of providing easy understanding. In FIG. **1D**, the arrows indicate the flow direction of the process gas.

[0099] As shown in FIG. **1D**, in the reactor **100**, the process gas flows into the second cells **24** and the third cells **25**, which do not have plugged portions on the inflow end face **20** side. The third cells **25** are filled with the functional material **50**, but the process gas can flow through gaps in the filled functional material **50**.

[0100] Since the second plugged portions **41** are provided on the outflow end face **21** side of the second cells **24**, the process gas that has flowed into the second cells **24** flows through the porous partition walls **30** into the third cells **25**, which are filled with the functional material **50**. For the process gas that has flowed into the third cells **25**, the trapping target gas is recovered (e.g., adsorbed) by the functional material **50**. Since the third plugged portions **42** are provided at the outflow end face **21** of the third cells **25**, the process gas that has recovered the trapping target gas by the functional material **50** flows into the first cells **23** through the porous partition walls **30**. The process gas that has flowed into the first cells **23** then flows out of the outflow end face **21**.

[0101] FIGS. **1A** to **1C** illustrate an example in which the third cells **25** are interposed entirely between the first cells **23** and the second cells **24**, but the third cells **25** may be interposed at least partially between the first cells **23** and the second cells **24** (i.e., there may be a portion(s) where the third cells **25** are not interposed, between the first cells **23** and the second cells **24**). Schematic views of the inflow end faces and outflow end faces of the reactor having such a structure are shown in FIGS. **2A** and **2B**. The cross section taken along the line b-b' in FIGS. **2A** and **2B** is the same as the cross section taken along the line a-a' in FIG. **1A**, and is therefore omitted.

[0102] As shown in FIGS. **2A** and **2B**, in the reactor **200**, all the third cells **25** are interposed between the first cells **23** and the second cells **24** in the direction of the line b-b', while all the third cells **25** are not interposed between the first cells **23** and the second cells **24** in the direction orthogonal to the line b-b'. Even with this structure of the reactor **200**, the above effects equivalent to those of the reactor **100** can be obtained.

[0103] The arrangement of the first cells **23**, the second cells **24**, and the third cells **25** is not limited to the forms illustrated in FIGS. **1A** to **1C** and FIGS. **2A** to **2B**, and may be any arrangement as long as the third cells **25** are interposed at least partially between the first cells **23** and the second cells **24**.

(1. Honeycomb Structure)

[0104] The shape of the honeycomb structure is not particularly limited as long as the honeycomb structure has the above structure. For example, the outer shape of the cross section orthogonal to the extending direction of the flow paths (the first cells **23**, the second cells **24**, and the third cells **25**) of the honeycomb structure can be a polygon such as a triangle, quadrangle, hexagon, or octagon, a circle, an ellipse, an oval, an egg shape, an elongated circular shape, a rounded quadrangle (a quadrangle consisting of a curved line as a whole, in which each side and corner is curved, and a radius of curvature of each side is greater than that of each corner). Among these, the shape of the honeycomb structure preferably has a rectangular outer shape of the cross section (i.e., the shape of the honeycomb structure is a quadrangular pillar shape). It should be noted that the end faces (inflow end face **20** and outflow end face **21**) have the same shape as the cross section.

[0105] In a preferable embodiment, the honeycomb structure has a quadrangular pillar shape with a length of one side of the inflow end face **20** and the outflow end face **21** of 100 to 500 mm (preferably 200 to 400 mm) and a length of the extending direction of the first cells **23**, the second cells **24**, and the third cells **25** of 100 to 1000 mm (preferably 300 to 500 mm). With the

honeycomb structure having this size, a sufficient amount of the functional material **50** filled in the third cells **25** can be ensured, so that practicality as a reactor **100**, **200** can be ensured.

[0106] The shape of each cell (the first cells **23**, the second cells **24** and the third cells **25**) is not particularly limited, but it may be polygon such as a triangle, quadrangle, hexagon, and octagon, a rounded quadrangle such as a circle, an ellipse, an oval, an egg shape, and an elongated circular shape, in the cross section of the honeycomb structure orthogonal to the extending direction of the flow paths (the first cells **23**, the second cells **25** and the third cells **25**). The shape of each cell may be alone or in combination of two or more. Among these shapes of each cell, a triangle, a quadrangle, a hexagon, an octagon, or a combination thereof is preferable. By providing each cell having such a shape, it is possible to reduce the pressure loss when the process gas flows. It should be noted that the shape of each cell in the cross section is the same as that of each cell at the end faces (inflow end face **20** and outflow end face **21**).

[0107] Here, examples of combinations of the respective cells having various shapes are shown in FIGS. **3** to **8**. FIGS. **3** to **8** illustrate partially enlarged views of the inflow end faces of the reactors including the respective cells having various shapes.

[0108] The embodiment of FIG. **3** has two types of hexagonal cells having different sizes. The first cells **23** and second cells **24** are hexagonal cells having the same size. The hexagonal third cells **25** are smaller than the hexagonal first cells **23** and second cells **24**.

[0109] The embodiment of FIG. **4** has each hexagonal cell having the same size.

[0110] The embodiment of FIG. **5** has two types of triangular cells having different sizes. The second cells **24** and third cells **25** are triangular cells having the same size. The triangular first cells **23** are larger than the triangular second cells **24** and third cells **25**.

[0111] Embodiment of FIG. **6** has octagonal first cells **23** and second cells **24**, and rectangular third cells **25**.

[0112] The embodiment of FIG. **7** has three types of rectangular cells having different sizes. The first cells **23** and the second cells **24** are rectangular cells having the same size. The third cells **25** has quadrangles having two types of sizes, which are smaller than the first cells **23** and the second cells **24**.

[0113] Embodiment of FIG. **8** has rectangular and hexagonal cells. The first cells **23** and the second cells **24** are rectangular cells having the same size. The third cells **25** are hexagonal cells.

[0114] In any of the embodiments of FIGS. **3-8**, the third cells **25** are positioned between the first cells **23** and the second cells **24**.

[0115] The honeycomb structure preferably has the first cells **23** and/or the second cells **24** positioned facing the outer peripheral wall **10**.

[0116] Here, a schematic view of the inflow end face of the reactor having this structure is illustrated in FIG. **9A**, a schematic view of the outflow end face is illustrated in FIG. **9B**, and a schematic view of the cross section taken along the line c-c' in FIGS. **9A** and **9A** is illustrated in FIG. **9C**.

[0117] As shown in FIGS. **9A-9C**, a reactor **300** is the same as the reactor **100** with the exception that the first cells **23** are positioned facing the outer peripheral wall **10**.

[0118] If the third cells **25** filled with the functional material **50** is located at the position facing the outer peripheral wall **10**, the inflow path or outflow path of the process gas is reduced, so that the process gas may not be sufficiently distributed in the third cells **25** facing the outer peripheral wall **10**, and the effect of the functional material **50** may not be sufficiently achieved. Therefore, as shown in FIGS. **9A-9C**, the first cells **23** can be placed at the position facing the outer peripheral wall **10**, thereby avoiding the use of wasteful functional material **50**, which would otherwise be difficult to produce the effect sufficiently, and thus reducing production costs. The effect can be obtained not only in the embodiment illustrated in FIGS. **9A-9C**, but also in the embodiment in which the second cells **24** are placed at the position facing the outer peripheral wall **10**, or in the embodiment in which both the first cells **23** and the second cells **24** are placed at the position facing

the outer peripheral wall **10**.

[0119] The honeycomb structure may be a honeycomb joined body that includes a plurality of honeycomb segments and joining layers that join outer peripheral surfaces (outer peripheral surfaces parallel to the extending direction of the honeycomb segment) of the plurality of honeycomb segments together. The use of the honeycomb joined body can increase the total cross-sectional area of the cells, which will be important for ensuring the flow rate of the process gas, while suppressing cracking. The joining layer can be formed by using a joining material. The joining material is not particularly limited, but a ceramic material obtained by adding a solvent such as water to form a paste can be used. The joining material may contain the same material as the outer peripheral wall **10** and the partition walls **30**. In addition to the role of joining the honeycomb segments to each other, the joining material can also be used as an outer peripheral coating material after joining the honeycomb segments.

[0120] The thickness of the partition walls **30** is not particularly limited, but from the viewpoints of ensuring the strength of the honeycomb structure and reducing the pressure loss when the process gas passes through the partition walls **30**, it may preferably be 0.05 mm to 5 mm, more preferably 0.10 mm to 4.5 mm, and even more preferably 0.15 mm to 4 mm.

[0121] Further, the “thickness of the partition walls **30**” as used herein refers to a length of a line segment across the partition wall **30** when the centers of gravity of adjacent cells are connected by the line segment in a cross section of the honeycomb structure orthogonal to the extending direction of the flow paths (the first cells **23**, the second cells **24** and the third cells **25**). The thickness of the partition walls **30** refers to an average value of the thicknesses of all the partition walls **30**.

[0122] The porosity of the partition walls **30** is not particularly limited, but from the viewpoints of ensuring the strength of the honeycomb structure and reducing the pressure loss when the process gas passes through the partition walls **30**, it may preferably be 30% or more and less than 80%, more preferably 35% to 75%, and even more preferably 40% to 70%.

[0123] It should be noted that the “porosity of the partition walls **30**” as used herein means a porosity measured using the mercury intrusion method in accordance with JIS R1655:2003.

[0124] The average pore diameter of the partition walls **30** is not particularly limited, but from the viewpoints of ensuring the strength of the honeycomb structure and reducing the pressure loss when the process gas passes through the partition walls **30**, it may preferably be 10 μm to 300 μm , more preferably 15 μm to 280 μm , and even more preferably 20 μm to 260 μm .

[0125] The “average pore diameter of the partition walls **30**” as used herein means the pore diameter of the partition walls **30** at 50% of an integrated value in a pore distribution determined by the mercury intrusion method in accordance with JIS R1655:2003.

[0126] The thickness of the outer peripheral wall **10** is not particularly limited, but from the viewpoints of ensuring the strength of the honeycomb structure and the like, it may preferably be 0.05 mm to 10 mm, more preferably 0.20 mm to 8 mm, and even more preferably 0.30 mm to 6 mm.

[0127] As used herein, the thickness of the outer peripheral wall **10** refers to a length, in a normal line direction of a side surface of the honeycomb structure, from a boundary between the outer peripheral wall **10** and the outermost cell or partition wall **30** to the outer peripheral surface of the honeycomb structure in the cross section of the honeycomb structure orthogonal to the extending direction of the flow paths (the first cells **23**, the second cells **24**, and the third cells **25**).

[0128] The cell density of the honeycomb structure is not particularly limited, but from the viewpoint of ensuring the strength of the honeycomb structure and increasing the amount of the functional material **50** filled, it may preferably be 0.05 cells/cm.^{sup.2} to 25 cells/cm.^{sup.2}, more preferably from 0.1 cells/cm.^{sup.2} to 20 cells/cm.^{sup.2}, and even more preferably 0.5 cells/cm.^{sup.2} to 15 cells/cm.^{sup.2}.

[0129] As used herein, the “cell density” refers a value obtained by dividing a number of cells by

an area of one end face (the total area of the partition walls **30**, the first cells **23**, the second cells **24** and the third cells **25**, excluding the outer peripheral wall **10**) of the honeycomb structure.

[0130] The materials of the outer peripheral wall **10** and the partition walls **30** are not particularly limited, but from the viewpoint of ensuring the strength of the honeycomb structure, they may contain one or more materials selected from cordierite, mullite, alumina, silicon carbide, and Si-bonded silicon carbide as main components.

[0131] The first plugged portions **40** and the second plugged portions **41** are preferably dense, although not particularly limited thereto. This configuration makes it easier for the process gas to flow through the flow paths as described above.

[0132] As used herein, the term “dense” means that the porosity is 10% or less. The porosity of the first plugged portions **40** and the second plugged portions **41** may be 5% or less.

[0133] The third plugged portions **42** are preferably porous, although not particularly limited thereto. As shown in FIG. **10**, such a configuration reduces the pressure loss when the process gas flows, because the process gas that has recovered the trapping target gas by the functional material **50** also flows out of the outflow end face **21** through the porous third plugged portions **42**. FIG. **10** is a partially enlarged view of the same cross section as FIG. **1D** to illustrate the flow of the process gas.

[0134] As used herein, the term “porous” means that the porosity is 20% or more. It should be noted that the porosity of the third plugged portions **42** may be 30% or more.

[0135] The materials of the first plugged portions **40**, the second plugged portions **41**, and the third plugged portions **42** are not limited and they can be made of known materials such as ceramics and resins. The characteristics of the first plugged portions **40**, the second plugged portions **41**, and the third plugged portions **42** can be ensured by selecting the type of material to be used, as needed.

[0136] For example, if the first plugged portions **40** and the second plugged portions **41** are dense, resin sheets, dense ceramics or glass can be selected and used, and the resin sheets are preferred in terms of production costs and productivity.

[0137] Similarly, if the third plugged portions **42** are porous, a resin porous sheet, porous ceramics or glass can be selected and used, the resin porous sheet is preferred in terms of production costs and productivity.

[0138] In addition to the first plugged portions **40**, the second plugged portions **41**, and the third plugged portions **42** described above, the honeycomb structure can further have fourth plugged portions at the third cells **25** on the inflow end face **20** side. Such a configuration can prevent the functional material **50** filled in the third cells **25** from falling out of the third cells **25**.

[0139] Here, a schematic view of the inflow end face of the reactor with the fourth plugged portions is shown in FIG. **11A**, and a schematic view of the cross section taken along the line d-d' in FIG. **11A** is shown in FIG. **11B**. The schematic view of the outflow end face according to this embodiment of the reactor is omitted because it is represented in the same manner as in FIG. **1B**.

[0140] As shown in FIGS. **11A** and **11bB**, a reactor **400** has the same basic structure as that of the reactor **100** as described above, but the former further has fourth plugged portions **43** at the third cells **25** on the inflow end face **20** side.

[0141] The material of the fourth plugged portions **43** is not limited and it can be made of ceramics, a resin or the like. The characteristics of the fourth plugged portions **43** can be ensured by selecting the type of material to be used, as needed. For example, the fourth plugged portions **43** may be dense or porous. However, if the fourth plugged portions **43** is dense, the pressure loss may increase because the process gas does not flow in from the inflow end face **20** of the third cells **25**. Therefore, the fourth plugged portions **43** are preferably porous. This configuration can ensure a flow path for the process gas as in FIG. **1D**, thus reducing the pressure loss when the process gas flows.

[0142] If the fourth plugged portions **43** is porous, a resin porous sheet, porous ceramics or glass can be selected and used, and the resin porous sheet is preferable in terms of production costs and

productivity.

(2. Functional Material)

[0143] The functional material **50** is not limited as long as it is capable of recovering the trapping target gas contained in the process gas, and for example, an adsorbent can be used. By using the adsorbent as the functional material **50**, the trapping target gas can be adsorbed and recovered, and the recovered gas can be easily desorbed by changing conditions such as temperatures.

[0144] The term “adsorbent” as used herein means a material capable of absorbing and storing the trapping target gas contained in the process gas.

[0145] The adsorbent may be appropriately selected according to the type of the trapping target gas, and is not particularly limited. Examples of the adsorbent effective for adsorption of the trapping target gas such as carbon dioxide (CO₂) include amine compounds, organometallic complexes, nanoporous ceramics or mesoporous silica on which the amine compounds and/or the organometallic complexes are supported. These can be used alone or in combination of two or more.

[0146] Non-limiting examples of the amine compounds include monoethanolamine (MEA), N-methyldiethanolamine (MDEA) and the like.

[0147] Non-limiting examples of the organometallic complexes include porous metal-organic frameworks (MOFs) that have a structure capable of adsorbing the trapping target gas in the pores.

[0148] The functional material **50** has a pellet shape. The use of the functional material **50** having the pellet shape can make it difficult for the functional material **50** to escape from the third cells **25**, and also increase the space between the functional materials **50**, thus lowering the pressure loss.

[0149] As used herein, the “pellet shape” means a shape having a substantially constant thickness, such as spherical, cylindrical, elliptical pillar-shaped, and polygonal pillar-shaped (e.g., triangular, quadrangular, pentagonal, hexagonal, etc.), the cross section of which is circular, elliptical, polygonal, or the like. The functional material **50** having the pellet shape has a particle diameter (which may hereinafter be referred to as a “pellet diameter”) of, for example, 0.025 to 10 mm, preferably 0.1 to 10 mm, more preferably 0.5 to 10 mm, and even more preferably 1 to 10 mm, although not limited thereto. The pellet diameter means an average value of the short and long diameters. A typical functional material **50** having the pellet shape is spherical or cylindrical material with a diameter or length of 0.3 to 10 mm.

[0150] The particle size of the functional material **50** having the pellet shape is not limited, but it may be smaller on the outflow end face **21** side than in the central portion in the extending direction of the third cells **25**. The process gas flowing from the second cells **24** and into the third cells **25** through the partition walls **30** flows more easily on the outflow end face **21** side than in the central portion in the extending direction of the third cells **25**. Therefore, the above configuration can suppress the preferential inflow of the process gas into the outflow end face **21** side and facilitate the uniform inflow of the process gas to the entire area in the extending direction of the third cells **25**.

[0151] The particle size of the functional material **50** having the pellet shape is determined by the following method: Using a micro video scope from Keyence Corporation, the diameter of the smallest circle of the circles circumscribed by the functional material **50** having the pellet shape is measured. The measurement is performed by randomly selecting **100** functional materials **50** having the pellet shape, and the average value is determined to be the particle diameter of the functional material **50** having the pellet shape.

[0152] The particle size of the functional material **50** having the pellet shape is preferably larger on the outer peripheral wall **10** side than in the central portion in the direction orthogonal to the extending direction of the third cells **25**. The central portion is easy for the process gas to flow, while the outer peripheral wall **10** side is more difficult for the process gas to flow than the central portion. Therefore, the above configuration can suppress the preferential inflow of the process gas into the third cells **25** in the central portion and facilitate the uniform inflow of the process gas into

the entire area in the direction orthogonal to the extending direction of the third cells **25**.

[0153] The filling rate of the functional material **50** having the pellet shape in the third cells **25** is not particularly limited, but it may preferably be 20 to 70%. By controlling the filling rate to such a level, it is possible to increase a recovery efficiency of the trapping target gas contained in the process gas. The filling rate can be measured, for example, by dividing the mass of the filled pellets (kg) by the density of the pellets (kg/m³) to calculate the volume of the filled pellets, and dividing it by the volume of the third cells **25** (m³) and multiplying it by 100.

[0154] The reactors **100**, **200**, **300** according to an embodiment of the invention are preferably provided so that each cell extends in the vertical direction, and the inflow end face **20** is at the top and the outflow end face **21** is at the bottom and are then used. By using the reactors **100**, **200**, and **300** in this manner, the functional material **50** can be prevented from escaping.

[0155] The reactor **400** according to an embodiment of the invention is provided with the fourth plugged portions **43**, so that there is no risk of the functional material **50** escaping. Therefore, the reactor **400** according to an embodiment of the invention can be provided so that each cell extends not only in the vertical direction, but also in various directions (e.g., horizontal).

<Method for Producing Reactor>

[0156] The method for producing the reactor according to an embodiment of the invention is not particularly limited as long as it is a method capable of making the structure as described above.

[0157] The method for producing the reactor **100**, **200**, **300** according to an embodiment of the invention includes: a step of preparing a honeycomb structure including: an outer peripheral wall **10** and porous partition walls **30** provided on an inner side of the outer peripheral wall **10**, the porous partition walls **30** defining first cells **23**, second cells **24**, and third cells **25** through which a process gas containing a trapping target gas can flow, each of the first cells **23**, the second cells **24** and the third cells **25** extending from an inflow end face **20** to an outflow end face **21** of the honeycomb structure, the third cells **25** being interposed between the first cells **23** and the second cells **24** (step **1A**); a step of forming second plugged portions **41** and third plugged portions **42** on the outflow end face **21** side of the second cells **24** and the third cells **25**, respectively (step **2A**); a step of filling the third cells **25** with a functional material **50** (step **3A**); and a step of forming first plugged portions **40** on the inflow end face **20** side of the first cells **23**.

[0158] In the step **1A**, the method for producing the honeycomb structure is not particularly limited, and it can be carried out in accordance with a known method in the art. For example, the honeycomb structure can be produced as follows: First, a green body containing ceramic powder is extruded into a desired shape to prepare a honeycomb formed body. At this time, the shape and density of each cell, and shapes and thicknesses of the outer peripheral wall **10** and the partition walls **30**, and the like, can be controlled by selecting dies and jigs in appropriate forms. As the ceramic powder, powder of the above-described ceramics or raw material powder that will become the above-described ceramics after firing (for example, cordierite-forming raw materials) can be used. The cordierite-forming raw materials are raw materials that will form cordierite by firing. The cordierite-forming raw materials may preferably have a chemical composition of 30 to 45% by mass of alumina (Al₂O₃) (including the proportion of aluminum hydroxide that is converted to alumina), 11 to 17% by mass of magnesia (MgO) and 42 to 57% by mass of silica (SiO₂). The green body can also contain a binder, a pore former, a dispersant, water, an organic solvent and the like. The porosity and average pore size of the partition walls **30** can be controlled by selecting the type and amount of the ceramic powder, binder, pore former and dispersant to be used, as needed.

[0159] The honeycomb formed body obtained above can be then dried and fired to obtain the honeycomb structure. Non-limiting examples of the drying method include known drying methods such as hot air drying, microwave drying, dielectric drying, drying under reduced pressure, drying in vacuum, and freeze drying. Among these, a drying method that combines the hot air drying with the microwave drying or dielectric drying is preferable because the entire honeycomb formed body

can be rapidly and uniformly dried.

[0160] Subsequently, in the step 2A, the second portions **41** and the third plugged portions **42** are formed on the outflow end face **21** side of the second cells **24** and the third cells **25**, respectively, of the honeycomb structure obtained above. The method for forming each plugged portion is not particularly limited, but it may be performed according to a known method. For example, when each plugged portion is made of a resin sheet, the second plugged portions **41** and the third plugged portions **42** can be formed by attaching the resin sheet to the outflow end faces **21** of the second cells **24** and the third cells **25**, respectively (step 4A). Further, when each plugged portion is made of ceramics, glass, or the like, first, a thin film having openings corresponding to the second cells **24** and the third cells **25** of the outflow end face **21** of the honeycomb structure where the second plugged portions **41** and the third plugged portion **42** are to be formed is attached. The outflow end face **21** of the honeycomb structure is then immersed in a slurry plugging material (ceramics, glass, etc.) and the plugging material can be allowed to penetrate into the second cells **24** and the third cells **25** of the honeycomb structure that are not plugged by the thin film, thereby forming the second plugged portions **41** and the third plugged portions **42**.

[0161] Subsequently, in the step 3A, the third cells **25** are filled with the functional material **50**. The filling method is not particularly limited, and it may be carried out in accordance with the conventional method.

[0162] Subsequently, in the step 4A, the first plugged portions **40** are formed on the inflow end face **20** side of the first cells **23**. The method for forming the first plugged portions **40** may be the same as the method for forming the second portions **41** and the third plugged portions **42** as described above.

[0163] The order of the steps 2A to 4A is not limited, with the exception that the step 3A is performed after the step 2A. For example, it can be carried out in the order of the steps 2A, 3A and 4A, or the steps 2A, 4A and 3A, or the steps 4A, 2A and 3A.

[0164] The method for producing the reactor **400** according to an embodiment of the invention includes: a step of preparing a honeycomb structure including: an outer peripheral wall **10** and porous partition walls **30** provided on an inner side of the outer peripheral wall **10**, the porous partition walls **30** defining first cells **23**, second cells **24**, and third cells **25** through which a process gas containing a trapping target gas can flow, each of the first cells **23**, the second cells **24** and the third cells **25** extending from an inflow end face **20** to an outflow end face **21** of the honeycomb structure, the third cells **25** being interposed between the first cells **23** and the second cells **24** (step 1B); a step of forming second plugged portions **41** and third plugged portions **42** on the outflow end face **21** side of the second cells **24** and the third cells **25**, respectively (step 2B); a step of filling the third cells **25** with a functional material **50** (step 3B); and a step of forming first plugged portions **40** and fourth plugged portions **43** on the inflow end faces **20** of the first cells **23** and the third cells **25**, respectively (step 4B).

[0165] The method for producing the honeycomb structure in the steps 1B to 3B can be carried out in the same manner as in the steps 1A to 3A.

[0166] In the step 4B, the first plugged portions **40** and the fourth plugged portions **43** are formed on the inflow end faces **20** of the first cells **23** and the third cells **25**, respectively. The method for forming the first plugged portions **40** and the fourth plugged portions **43** on the inflow end faces **20** of the first cells **23** and third cells **25** may be the same as the method for forming each plugged portion as described above.

[0167] The steps 2B to 4B are carried out in the order of the steps 2B, 3B and 4B.

<Gas Recovery Device>

[0168] The gas recovery device according to an embodiment of the invention can be suitably used to recover and release the trapping target gas contained in the process gas.

[0169] FIG. **12** is a schematic view of the gas recovery device according to an embodiment of the invention.

[0170] As shown in FIG. 12, a gas recovery device **1000** includes: a reactor **1100**; a gas feed pipe **1200** configured to feed the process gas or a purge gas to an inflow port **1110** of the reactor **1100**; and a gas discharge pipe **1300** configured to discharge the process gas or the purge gas from an outflow port **1120** of the reactor **1100**. The reactors **100**, **200**, **300**, and **400** as described above are used for the reactor **1100**, which can increase the recovery amount of the trapping target gas by increasing the amount of the functional material **50** retained while reducing the increase in pressure loss. Therefore, the gas recovery device **1000** can also increase the recovery amount of the trapping target gas while suppressing the increase in pressure loss.

[0171] The gas feed pipe **1200** has a gas feed branch pipe that branches into two parts. The gas feed branch pipe can be a first gas feed branch pipe **1210** configured to feed the process gas and a second gas feed branch pipe **1220** configured to feed the purge gas.

[0172] The gas discharge pipe **1300** has a gas discharge branch pipe that branching into two parts. The gas discharge branch pipe can be a first gas discharge branch pipe **1310** configured to discharge the process gas and a second gas discharge branch pipe **1320** configured to discharge the purge gas.

[0173] The gas recovery device **1000** further includes a feed gas switching valve **1400** configured to shut off the first gas feed branch pipe **1210** or the second gas feed branch pipe **1220**, and a discharge gas switching valve **1500** configured to shut off the first gas discharge branch pipe **1310** or the second gas discharge branch pipe **1320**.

[0174] In the gas recovery device **1000** having the above structure, when recovering the trapping target gas contained in the process gas, the feed gas switching valve **1400** is switched to shut off the second gas feed branch pipe **1220** and open the first gas feed branch pipe **1210**, and the discharge gas switching valve **1500** is switched to shut off the second gas discharge branch pipe **1320** and open the first gas discharge branch pipe **1310**. The process gas containing the trapping target gas is then fed from the first gas feed branch pipe **1210** to the inflow port **1110** of the reactor **1100** via the gas feed pipe **1200**. The process gas fed to the reactor **1100** is discharged from the outflow port **1120** after the trapping target gas is recovered. The discharged process gas is discharged from the first gas discharge branch pipe **1310** via the gas discharge pipe **1300**.

[0175] Subsequently, when the trapping target gas that has been recovered in the reactor **1100**, the feed gas switching valve **1400** is switched to open the second gas feed branch pipe **1220** and shut off the first gas feed branch pipe **1210**, and the discharge gas switching valve **1500** is switched to open the second gas discharge branch pipe **1320** and shut off the first gas discharge branch pipe **1310**. The purge gas is then fed from the second gas feed branch pipe **1220** to the inflow port **1110** of the reactor **1100** via the gas feed pipe **1200**. The purge gas fed to the reactor **1100** is discharged from the outflow port **1120** together with the trapping target gas captured in the functional material **50** of the reactor **1100**. The purge gas containing the trapping target gas is discharged from the second gas discharge branch pipe **1320** via the gas discharge pipe **1300**.

[0176] As used herein, the “purge gas” means a gas that can desorb the trapping target gas captured in the functional material **50** of the reactor **1100** and discharge it from the reactor **1100**. The purge gas may be appropriately selected depending on the type of the trapping target gas. For example, when the trapping target gas is carbon dioxide, water vapor or the like may be used. The water vapor may preferably be at an elevated temperature such as 100° C. or higher (e.g., 120° C.).

[0177] To heat the purge gas to a predetermined temperature, the gas recovery device **1000** may further include a heating mechanism configured to heat the reactor **1100**. With this configuration, the purge gas at room temperature can be fed from the second gas feed branch pipe **1220** to heat the purge gas to the predetermined temperature in the reactor **1100**. Therefore, there is no need to preheat the purge gas fed to the gas recovery device **1000**.

[0178] FIG. 13 is a schematic view of a structure of a gas recovery device according to another embodiment of the invention.

[0179] As shown in FIG. 13, a gas recovery device **2000** has the same basic structure as that of the

gas recovery device **2000** in FIG. **12**. In other words, the gas recovery device **2000** includes: a reactor **1100**; a gas feed pipe **1200** configured to feed a process gas or a purge gas to an inflow port **1110** of the reactor **1100**; and a gas discharge pipe **1300** configured to discharge the process gas or the purge gas from an outflow port **1120** of the reactor **1100**. Therefore, the gas recovery device **2000** can also increase the recovery amount of the trapping target gas while suppressing the increase in pressure loss.

[0180] The gas feed pipe **1200** has two independent gas feed pipes. The two independent gas feed pipes are a first gas feed pipe **2100** configured to feed the process gas, and a second gas feed pipe **2200** configured to feed the purge gas. The gas discharge pipe **1300** has two independent gas discharge pipes. The two independent gas discharge pipes are a first gas discharge pipe **2300** configured to discharge the process gas, and a second gas discharge pipe **2400** configured to discharge the purge gas.

[0181] The reactor **1100** can be located between the first gas feed pipe **2100** and the first gas discharge pipe **2300** or between the second gas feed pipe **2200** and the second gas discharge pipe **2400**.

[0182] The gas recovery device **2000** further includes a transfer mechanism (not shown) configured to transfer the reactor **1100** between the first gas feed pipe **2100** and the first gas discharge pipe **2300** or between the second gas feed pipe **2200** and the second gas discharge pipe **2400**. The transfer mechanism is not particularly limited, and a known transfer mechanism (for example, a transfer mechanism having a motor drive) can be used.

[0183] In the gas recovery device **2000** having the above structure, when recovering the trapping target gas contained in the process gas, the reactor **1100** is placed between the first gas feed pipe **2100** and the first gas discharge pipe **2300** by the transfer mechanism. The process gas containing the trapping target gas is then fed from the first gas feed pipe **2100** to the inflow port **1110** of the reactor **1100**. The process gas fed to the reactor **1100** is discharged from the outflow port **1120** after the trapping target gas is recovered. The discharged process gas is discharged from the first gas discharge pipe **2300**.

[0184] Subsequently, when the trapping target gas recovered in the reactor **1100** is released, the reactor **1100** is placed between the second gas feed pipe **2200** and the second gas discharge pipe **2400** by the transfer mechanism. The purge gas is then fed from the second gas feed pipe **2200** to the inflow port **1110** of the reactor **1100**. The purge gas fed to the reactor **1100** is discharged from the outflow port **1120** together with the trapping target gas captured in the functional material **50** of the reactor **1100**. The purge gas containing the trapping target gas is discharged from the second gas discharge pipe **2400**. The same purge gas as described above can be used.

[0185] To heat the purge gas to a predetermined temperature, the gas recovery device **2000** may further include a heating mechanism configured to heat the reactor **1100** located between the second gas feed pipe **2200** and the second gas discharge pipe **2400**. With this configuration, the purge gas at room temperature can be fed from the second gas feed pipe **2200** to heat the purge gas to the predetermined temperature in the reactor **1100**. Therefore, there is no need to preheat the purge gas fed to the gas recovery device **2000**.

<Gas Recovery System>

[0186] The gas recovery system according to an embodiment of the invention can be suitably used to recover and release the trapping target gas contained in the process gas.

[0187] FIG. **14** is a schematic view of a structure of a gas recovery system according to an embodiment of the invention.

[0188] As shown in FIG. **14**, a gas recovery system **3000** includes a gas recovery device **3100**, a gas release device **3200**, and a transfer device **3300**.

[0189] The gas recovery device **3100** includes: an attachable/detachable portion **3110** configured to attach and detach the reactor **1100**; a gas feed pipe **1200** configured to feed the process gas to the inflow port **1110** of the reactor **1100**; and a gas discharge pipe **1300** configured to discharge the

process gas from the outflow port **1120** of the reactor **1100**.

[0190] The gas release device **3200** includes; an attachable/detachable portion **3210** configured to attach and detach the reactor **1100**; a gas feed pipe **1200** configured to feed the purge gas to the inflow port **1110** of the reactor **1100**; and a gas discharge pipe **1300** configured to discharge the purge gas from the outflow port **1120** of the reactor **1100**.

[0191] The transfer device **3300** can transfer the reactor **1100** that has recovered the trapping target gas by the gas recovery device **3100** to the gas release device **3200**, and can also transfer the reactor **1100** that has released the trapping target gas by the gas release device **3200** to the gas recovery device **3100**.

[0192] In the gas recovery system **3000** having the structure as described above, when the trapping target gas contained in the process gas is recovered, the reactor **1100** is placed at the attachable/detachable portion **3110** of the gas recovery device **3100** by the transfer device **3300**. The process gas containing the trapping target gas is then fed from the gas feed pipe **1200** of the gas recovery device **3100** to the inflow port **1110** of the reactor **1100**. The process gas fed to the reactor **1100** is discharged from the outflow port **1120** after the trapping target gas is recovered. The discharged process gas is discharged from the gas discharge pipe **1300**.

[0193] Subsequently, when the trapping target gas recovered in the reactor **1100** is released, the reactor **1100** placed in the gas recovery device **3100** is transferred to the attachable/detachable portion **3210** of the gas release device **3200** by the transfer device **3300**. The purge gas is then fed from the gas feed pipe **1200** of the gas release device **3200** to the inflow port **1110** of the reactor **1100**. The purge gas fed to the reactor **1100** is discharged from the outflow port **1120** together with the trapping target gas captured in the functional material **50** of the reactor **1100**. The purge gas containing the trapping target gas is discharged from the gas discharge pipe **1300**. The same purge gas as described above can be used.

[0194] To heat the purge gas to a predetermined temperature, the gas release device **3200** may further include a heating mechanism configured to heat the reactor **1100**. With this configuration, the purge gas at room temperature can be fed from gas feed pipe **1200** and the purge gas can be heated to the predetermined temperature in the reactor **1100**. Therefore, there is no need to preheat the purge gas fed to the gas release device **3200**.

[0195] The transfer device **3300** is not particularly limited, but it may include a vehicle. This configuration allows the reactor **1100** to be transferred efficiently even if the gas recovery device **3100** and the gas release device **3200** are in separated positions.

DESCRIPTION OF REFERENCE NUMERALS

[0196] **10** outer peripheral wall [0197] **20** inflow end face [0198] **21** outflow end face [0199] **23** first cell [0200] **24** second cell [0201] **25** third cell [0202] **30** partition wall [0203] **40** first plugged portion [0204] **41** second plugged portion [0205] **42** third plugged portion [0206] **43** fourth plugged portion [0207] **50** functional material [0208] **100, 200, 300, 400** reactor [0209] **1000, 2000** gas recovery device [0210] **1100** reactor [0211] **1110** inflow port [0212] **1120** outflow port [0213] **1200** gas feed pipe [0214] **1210** first gas feed branch pipe [0215] **1220** second gas feed branch pipe [0216] **1300** gas discharge pipe [0217] **1310** first gas discharge branch pipe [0218] **1320** second gas discharge branch pipe [0219] **1400** feed gas switching valve [0220] **1500** discharge gas switching valve [0221] **2100** first gas feed pipe [0222] **2200** second gas feed pipe [0223] **2300** first gas discharge pipe [0224] **2400** second gas discharge pipe [0225] **3000** gas recovery system [0226] **3100** gas recovery device [0227] **3110** attachable/detachable portion [0228] **3200** gas release device [0229] **3210** attachable/detachable portion [0230] **3300** transfer device

Claims

1. A reactor, comprising: a honeycomb structure comprising: an outer peripheral wall; porous partition walls provided on an inner side of the outer peripheral wall, the partition walls defining

first cells, second cells and third cells through which a process gas containing a trapping target gas can flow, each of the first cells, the second cells and the third cells extending from an inflow end face to an outflow end face of the honeycomb structure; first plugged portions provided at the first cells on the inflow end face side; second plugged portions provided at the second cells on the outflow end face side; and third plugged portions provided at the third cells on the outflow end face side, the third cells being interposed between the first cells and the second cells; and a functional material having a pellet shape which has a particle diameter of 0.025 to 10 mm, the functional material being filled in the third cells.

2. The reactor according to claim 1, wherein the honeycomb structure further comprises fourth plugged portions provided at the third cells on the inflow end face side.

3. The reactor according to claim 1, wherein the third plugged portions are porous.

4. The reactor according to claim 1, wherein the first plugged portions and the second plugged portions are dense.

5. The reactor according to claim 2, wherein the third plugged portions are porous.

6. The reactor according to claim 1, wherein each of the first cells, the second cells, and the third cells has a triangular, quadrangular, hexagonal, or octagonal shape, or a combination thereof, in a cross section orthogonal to an extending direction of the first cells, the second cells, and the third cells.

7. The reactor according to claim 1, wherein the outer peripheral wall and the partition walls comprise at least one selected from cordierite, mullite, alumina, silicon carbide, and Si-bonded silicon carbide as a main component.

8. The reactor according to claim 1, wherein the partition walls have a thickness of 0.05 mm to 5 mm.

9. The reactor according to claim 1, wherein the partition walls have a porosity of 30% or more and less than 80%.

10. The reactor according to claim 1, wherein the partition walls have an average pore diameter of 10 μm to 300 μm .

11. The reactor according to claim 1, wherein the functional material is an adsorbent.

12. The reactor according to claim 11, wherein the adsorbent is at least one selected from an amine compound, an organometallic complex, and a nanoporous ceramic or mesoporous silica on which the amine compound and/or the organometallic complex are supported.

13. The reactor according to claim 12, wherein a particle diameter of the functional material having the pellet shape is lower on the outflow end face side than in a central portion in the extending direction of the third cells.

14. The reactor according to claim 12, wherein the particle diameter of the functional material having the pellet shape is higher on the outer peripheral wall side than in the central portion in a direction orthogonal to the extension direction of the third cells.

15. The reactor according to claim 1, wherein the first cells and/or the second cells are located at positions facing the outer peripheral wall.

16. The reactor according to claim 1, wherein the first plugged portions, the second plugged portions, and the third plugged portions are made of a resin sheet.

17. The reactor according to claim 2, wherein the third plugged portions and the fourth plugged portions are made of a porous resin sheet.

18. The reactor according to claim 1, wherein the honeycomb structure has a quadrangular pillar shape having a length of one side of the inflow end face and the outflow end face of 100 to 500 mm, and a length in the extending direction of the first cells, the second cells, and the third cells of 100 to 1,000 mm.

19. A method for producing a reactor, the method comprising the steps of: preparing a honeycomb structure comprising: an outer peripheral wall and porous partition walls provided on an inner side of the outer peripheral wall, the porous partition walls defining first cells, second cells, and third

cells through which a process gas containing a trapping target gas can flow, each of the first cells, the second cells and the third cells extending from an inflow end face to an outflow end face of the honeycomb structure, the third cells being interposed between the first cells and the second cells; forming second plugged portions and third plugged portions on the outflow end face side of the second cells and the third cells, respectively; filling the third cells with a functional material having a pellet shape which has a particle diameter of 0.025 to 10 mm; and forming first plugged portions on the inflow end face side of the first cells.

20. A method for producing a reactor, the method comprising the steps of: preparing a honeycomb structure comprising: an outer peripheral wall and porous partition walls provided on an inner side of the outer peripheral wall, the porous partition walls defining first cells, second cells, and third cells through which a process gas containing a trapping target gas can flow, each of the first cells, the second cells and the third cells extending from an inflow end face to an outflow end face, the third cells being interposed between the first cells and the second cells; forming second plugged portions and third plugged portions on the outflow end face side of the second cells and the third cells, respectively; filling the third cells with a functional material having a pellet shape which has a particle diameter of 0.025 to 10 mm; and forming first plugged portions and fourth plugged portions on the inflow end faces of the first cells and the third cells, respectively.

21. A gas recovery device for recovering and releasing a trapping target gas contained in a process gas, the gas recovery device comprising: the reactor according to claim 1; a gas feed pipe configured to feed the process gas or a purge gas to an inflow port of the reactor; and a gas discharge pipe configured to discharge the process gas or the purge gas from an outflow port of the reactor.

22. The gas recovery device according to claim 21, wherein the gas feed pipe has a gas feed branch pipe that branches into two parts, the gas feed branch pipe being a first gas feed branch pipe configured to feed the process gas and a second gas feed branch pipe configured to feed the purge gas, wherein the gas discharge pipe has a gas discharge branch pipe that branches into two portions, the gas discharge branch pipe being a first gas discharge branch pipe configured to discharge the process gas and a second gas discharge branch pipe configured to discharge the purge gas, and wherein the gas recovery device further comprises: a feed gas switching valve configured to shut off the first gas feed branch pipe or the second gas feed branch pipe; and a discharge gas switching valve configured to shut off the first gas discharge branch pipe or the second gas discharge branch pipe.

23. The gas recovery device according to claim 21, wherein the gas feed pipe has two independent gas feed pipes, the gas feed pipes being a first gas feed pipe configured to feed the process gas and a second gas feed pipe configured to feed the purge gas, wherein the gas discharge pipe has two independent gas discharge pipes, the gas discharge pipes being a first gas discharge pipe configured to discharge the process gas and a second gas discharge pipe configured to discharge the purge gas, wherein the reactor can be provided between the first gas feed pipe and the first gas discharge pipe or between the second gas feed pipe and the second gas discharge pipe, and wherein the gas recovery device further comprises a transfer mechanism configured to transfer the reactor between the first gas feed pipe and the first gas discharge pipe or between the second gas feed pipe and the second gas discharge pipe.

24. The gas recovery device according to claim 21, further comprising a heating mechanism configured to heat the reactor.

25. A gas recovery system for recovering and releasing a trapping target gas contained in a process gas, the gas recovery system comprising: a gas recovery device comprising an attachable/detachable portion configured to attach and detach the reactor according to claim 1, a gas feed pipe configured to feed the process gas to an inflow port of the reactor, and a gas discharge pipe configured to discharge the process gas from an outflow port of the reactor; a gas release device comprising an attachable/detachable portion configured to attach and detach the reactor, a gas

feed pipe configured to feed a purge gas to the inflow port of the reactor, and a gas discharge pipe configured to discharge the purge gas from the outflow port of the reactor; and a transfer device configured to transfer the reactor that has recovered the trapping target gas by the gas recovery device to the gas release device and also to transfer the reactor that has released the trapping target gas by the gas release device to the gas recovery device.

26. The gas recovery system according to claim 25, wherein the transfer device comprises a vehicle.
